

Assignment:-4

Task 1:- List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

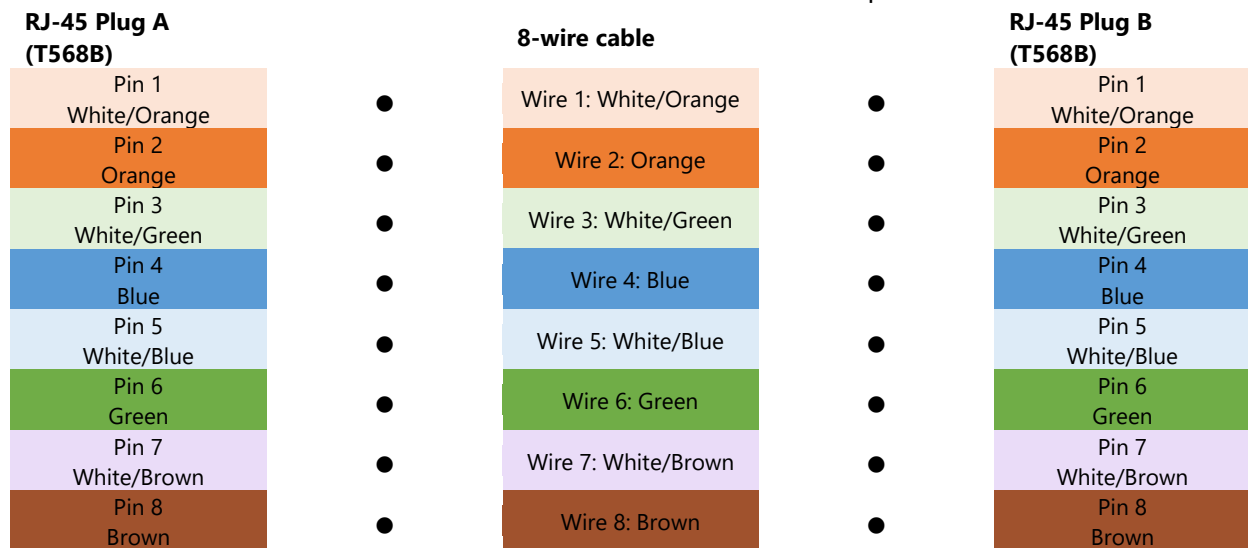
Basis	Ethernet Cable	Fiber Optic Cable
1. Transmission Medium	Uses copper wires to transmit electrical signals.	Uses glass or plastic fibers to transmit light signals.
2. Speed & Distance	Suitable for short distances, commonly up to 100 meters.	Supports very high speeds and much longer distances.
3. Interference	Can be affected by electromagnetic interference (EMI).	Resistant to electromagnetic interference.

Real-World Examples

- **Ethernet Cable:** Connecting a desktop computer to a home Wi-Fi router or network switch.
- **Fiber Optic Cable:** Connecting buildings in a college campus backbone or linking servers in a data center.

Task 2:- Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

Both RJ-45 connectors use the same T568B pinout.



Task 3:- Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

How to Attach an RJ45 Connector to a CAT6 Cable

Things Needed

- CAT6 cable
- RJ45 connector
- Crimping tool
- Cable cutter
- Cable tester

Steps

1. Cut the CAT6 cable to the needed length.
2. Remove the outer cover of the cable using a cable stripper.
3. Take out the 8 small wires inside the cable.
4. Straighten the wires with your fingers.
5. Put the wires in T568B order:
 1. White/Orange
 2. Orange
 3. White/Green
 4. Blue
 5. White/Blue
 6. Green
 7. White/Brown
 8. Brown
6. Cut the wire ends so they are the same length.
7. Put all 8 wires into the RJ45 connector. Make sure the wires are in the correct order.
8. Check the connector. Make sure all 8 wires go to the end of the connector.
9. Put the connector into the crimping tool and press it firmly.
10. Do the same thing on the other end of the cable. Use the same T568B order.
11. Test the cable with a cable tester. If the wires are correct, the cable is ready to use.

Result

The RJ45 connector is attached to the CAT6 cable successfully using the T568B standard.

Task 4:- Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Consider speed, distance, and budget for your answer.

For a gaming cafe with 10 PCs, I would use Ethernet cables with RJ45 connectors. Ethernet is fast enough for LAN gaming and works well for the short distance between PCs. It is also cheaper than fiber optic cable, so it is better for a small café budget. I would use an RJ45 crimper because it is simple and easy to use for making Ethernet cables.